

When $\beta \gg 1$, in general the hydrodynamic forces dominate the system, and in contrast when $\beta \ll 1$ the magnetic forces are expected to dominate the system. It is important to note that because gravity scales differently with length than these other forces, even in a low β system gravity can be important. The magnetic RT instability is just such an example where this can happen.

When magnetic diffusion (η_D) is included, the magnetic field and the flow are no longer frozen in to each other. Eq. (75) then becomes

$$\frac{\partial \mathbf{B}}{\partial t} - \nabla \times (\mathbf{u} \times \mathbf{B} - \mu_0 \eta_D \mathbf{J}) = 0. \quad (83)$$

Including diffusion in the induction equation also result in the addition of the Ohmic heating term, taking the form $\mu_0 \eta_D \mathbf{J}^2$. to the RHS of Eq. (74).

Since the diffusion term in Eq. (83) is in effect a second-order differential operator on $\mathbf{B}$, it is more effective at smaller scales. The importance of diffusion at a given scale can be understood by calculating the magnetic Reynolds number (the ratio of the diffusion time to the advection time) given by

$$R_m = \frac{VL}{\eta_D}, \quad (84)$$

where V is a characteristic velocity of the system, and L a characteristic length scale. In a collisional plasma the magnitude of η_D relates to the level the current is being reduced as a result of the current carriers (normally electrons) colliding with other particles.

Using an ideal MHD system (no viscosity or diffusivity) and taking possibly the simplest situation of an isotropic, homogeneous plasma with density ρ at rest that is permeated by a uniform, unidirectional magnetic field with no gravity, we simply determine the wave spectra of this system. However, before performing any analysis it is common to absorb the factor of $1/\sqrt{\mu_0}$ into the $\mathbf{B}$ term seen in Eq. (73), leading to

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) + \nabla p - (\nabla \times \mathbf{B}) \times \mathbf{B} + \rho \mathbf{g} = 0. \quad (85)$$

It is worth noting that if one is using the CGS unit system, absorbing the factor of $1/\sqrt{4\pi}$ into the $\mathbf{B}$ results in exactly the same equation.

In the situation we are studying, unlike hydrodynamics where sound waves are the only waves that exist, there are three kinds of waves in the system. Firstly there is the Alfvén wave, which has frequency

$$\omega^2 = k_x^2 \frac{B_{x,0}^2}{\rho} \quad (86)$$

where we have assumed initially we have a uniform magnetic field that is in the x direction and of magnitude $B_{x,0}$, and with k_x the component of the wave vector along the magnetic field. These waves are incompressible and propagate along the magnetic field at phase speed $V_A = B_{x,0}/\sqrt{\rho}$.

Then there are two compressible waves, the slow and fast-mode magnetoacoustic waves, which have frequencies

$$\omega^2 = \frac{1}{2} k^2 (C_s^2 + V_A^2) \left(1 \pm \sqrt{1 - \frac{4C_s^2 V_A^2 \cos^2 \theta}{(C_s^2 + V_A^2)^2}} \right), \quad (87)$$

where θ is the angle between the wave vector $\mathbf{k}$ and $\mathbf{B}$, k is the magnitude of $\mathbf{k}$, and C_s is the sound speed. The solution with the $+$ sign is the fast mode, and that with the $-$ sign is the slow mode. This means that fast mode waves propagate at the largest speed in the direction across the magnetic field, while slow mode waves propagate most quickly along it. Since ω is a linear function of k , these waves are not dispersive. More information on the MHD wave spectra can be found in [18], for example.

12. The magnetic Rayleigh–Taylor instability

The previous section contained a discussion of the coupling of a magnetic field, through the framework of MHD, to the hydrodynamic body forces leading to an additional body force, namely the Lorentz force. In this section, we will explore the effect this additional force, and the requirement to solve for the evolution of the magnetic field under the constraint that the field has zero divergence, has on the Rayleigh–Taylor instability.

12.1. Studying the magnetic Rayleigh–Taylor instability in an incompressible 2D model

In this subsection we present some of the key ideas and results needed to understand how the presence of magnetic fields alters RT. Here we will focus on cases where the magnetic field vector is perpendicular to the direction of gravity. This instability was first investigated by Kruskal and Schwarzschild [404]. The case with the magnetic field parallel to gravity, and other interesting examples, are presented in Chandrasekhar [405].

Our main investigation will focus on a simplified setting of the linear stability of a discontinuous density inversion with a uniform, horizontal magnetic field under gravity in 2D. The initial conditions we study in this section are

$$\rho_0 = \begin{cases} \rho_1 & \text{if } z < 0 \\ \rho_2 & \text{if } z > 0, \end{cases} \quad (88)$$

$$u_{x,0} = 0, \quad u_{z,0} = 0, \quad (89)$$

$$B_{x,0} = B_x, \quad B_{z,0} = 0, \quad (90)$$

$$p_0(z) = p_0(0) - \rho_0 g z. \quad (91)$$

We also assume the system is inviscid, ideal and incompressible (i.e. we solve the evolution using Eqs. (72), (75), (76), (85) and $\nabla \cdot \mathbf{u} = 0$, the last of which removes the requirement to solve for the evolution of the gas pressure).

These assumptions and initial conditions result in the following linear equations for the perturbed quantities $\delta \rho$, $\delta \mathbf{u}$ and $\delta \mathbf{B}$

$$\frac{\partial \delta \rho}{\partial t} = -\delta \mathbf{u} \cdot \nabla \rho_0, \quad (92)$$

$$\rho_0 \frac{\partial \delta \mathbf{u}}{\partial t} = -\nabla \delta p + (\nabla \times \delta \mathbf{B}) \times \mathbf{B}_0 + \delta \rho \mathbf{g}, \quad (93)$$

$$\frac{\partial \delta \mathbf{B}}{\partial t} = \nabla \times (\delta \mathbf{u} \times \mathbf{B}_0), \quad (94)$$

$$\nabla \cdot \delta \mathbf{u} = \nabla \cdot \delta \mathbf{B} = 0. \quad (95)$$

12.1.1. Lagrangian displacements in linear MHD

There are several ways in which the above equations can be manipulated to allow for further investigation of the ideal MHD instabilities. Here, we will apply the use of the Lagrangian displacement vector field, since it is an effective way of reducing the equations to a meaningful form.

The Lagrangian displacement, denoted by the vector ξ , describes the difference in position of a fluid element at a time t in two different flows where the fluid element in each flow is at the same position $\mathbf{a}$ at time $t = 0$ (see Fig. 41). Taking the two flows to be the initial flows $\mathbf{u} = \mathbf{u}_0 = \mathbf{0}$ and the perturbed flows $\mathbf{u} = \mathbf{u}_0 + \delta \mathbf{u} = \delta \mathbf{u}$, then ξ describes the manner in which the instability displaces fluid elements. The change of ξ with time must then be

$$\delta \mathbf{u} = \frac{\partial \xi}{\partial t}. \quad (96)$$

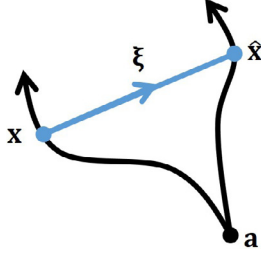


Fig. 41. The schematic diagram of the Lagrangian displacement ξ for two flows; one which takes the fluid parcel from position $\mathbf{a}$ to position $\mathbf{x}$ at time t and one which takes it to $\hat{\mathbf{x}}$.

For a more detailed explanation of Lagrangian displacements see, for example, Hillier [406].

We can insert this into Eqs. (92) and (94), and (with the appropriate simplifications and choice of integration constants) find

$$\delta\rho = -\xi \cdot \nabla\rho_0, \quad (97)$$

$$\delta\mathbf{B} = \nabla \times (\xi \times \mathbf{B}_0). \quad (98)$$

This means our linear equations become

$$\rho_0 \frac{\partial^2 \xi}{\partial t^2} = -\nabla\delta p + (\nabla \times \nabla \times (\xi \times \mathbf{B}_0)) \times \mathbf{B}_0 - (\xi \cdot \nabla\rho_0)\mathbf{g}, \quad (99)$$

$$\nabla \cdot \xi = 0, \quad (100)$$

as we also note that due to the form of Eq. (98) the solenoidal condition on the magnetic field is now automatically satisfied. Later, we will use Eq. (100) to remove the pressure from Eq. (99). Once this is done, what we have achieved is writing Eq. (99) in the form

$$\rho_0 \frac{\partial^2 \xi}{\partial t^2} = \mathbf{F}(\xi), \quad (101)$$

i.e. the second derivative of ξ with respect to time (the acceleration) relates to a force operator $\mathbf{F}$ that only depends on ξ . This now gives us quite a general tool to investigate instabilities (though not one we will apply here). If for a given ξ , we have $\xi \cdot \mathbf{F}(\xi)$ greater than zero somewhere, then the force in the system is working to increase the displacement (i.e. we have instability). Otherwise, the system is stable.

12.1.2. Derivation of the dispersion relation

Because of the initial conditions prescribed above, there is no variation in the equilibrium in the horizontal direction. This means that in these directions, a normal mode decomposition can be used, so that perturbations (both δp and the components of ξ) are taken to be of the form

$$\delta f(\mathbf{x}, t) = \tilde{f}(z) \exp(i\mathbf{k} \cdot \mathbf{x} - i\omega t). \quad (102)$$

This results in the Lagrangian displacement given by

$$-i\omega\xi = \delta\mathbf{u}. \quad (103)$$

In addition, this gives the following set of linear equations (dropping the $\sim$ s from the eigenfunctions)

$$-\rho_0\omega^2\xi_x = -ik\delta p, \quad (104)$$

$$-\rho_0\omega^2\xi_z = -\frac{\partial}{\partial z}\delta p + B_{x,0}\left(\frac{\partial^2}{\partial z^2} - k^2\right)\xi_z + \xi_z\frac{\partial}{\partial z}\rho_0 g. \quad (105)$$

We also have the incompressible condition

$$ik\xi_x = -\frac{\partial}{\partial z}\xi_z. \quad (106)$$

Taking the x derivative of Eq. (104) allows ξ_x to be eliminated from the equations,

$$-\rho_0\omega^2(ik\xi_x) = \rho_0\omega^2\frac{\partial}{\partial z}\xi_z = k^2\delta p. \quad (107)$$

Rearranging Eq. (107) and differentiating with respect to z gives

$$\frac{\partial}{\partial z}\delta p = \frac{1}{k^2}\frac{\partial}{\partial z}\left(\rho_0\omega^2\frac{\partial}{\partial z}\xi_z\right). \quad (108)$$

On substitution into Eq. (105) we obtain

$$\begin{aligned} \rho_0\omega^2\xi_z &= \frac{1}{k^2}\frac{\partial}{\partial z}\left((\rho_0\omega^2 - (kB_{x,0})^2)\frac{\partial}{\partial z}\xi_z\right) \\ &\quad + (kB_{x,0})^2\xi_z - \xi_z g \frac{d}{dz}\rho_0. \end{aligned} \quad (109)$$

Using this equation, we will now determine the variation of ξ_z with z and the dispersion relation.

Away from $z = 0$, Eq. (109) simplifies to

$$(\rho_0\omega^2 - (kB_{x,0})^2)\left(\frac{\partial^2}{\partial z^2} - k^2\right)\xi_z = 0. \quad (110)$$

In the following, we will assume the non-trivial case: $\rho_0\omega^2 - (kB_{x,0})^2 \neq 0$. Therefore, Eq. (110) reduces to

$$\left(\frac{\partial^2}{\partial z^2} - k^2\right)\xi_z = 0, \quad (111)$$

to which the solutions are

$$\xi_z(z) = C_0 \exp(-kz) + C_1 \exp(kz). \quad (112)$$

To maintain a finite energy in the system we require that the as $z \rightarrow \pm\infty$, $\xi_z(z) \rightarrow 0$. Combining this condition with the continuity of ξ_z at $z = 0$ leads to the solution

$$\xi_z(z) = C_0 \begin{cases} \exp(kz) & \text{if } z < 0, \\ \exp(-kz) & \text{if } z > 0. \end{cases} \quad (113)$$

With the full spatial distribution of ξ_z , ω can be determined. The next step is to integrate Eq. (109) from $-\epsilon$ to ϵ and take the limit $\epsilon \rightarrow 0$. This means that any terms of this equation that are continuous or form a step function at $z = 0$ will not contribute to the integration in the limit $\epsilon \rightarrow 0$. Therefore, we are only interested in terms containing the derivatives of step functions (i.e. the first and third terms of the right-hand-side of Eq. (109)). Upon integration (and evaluation at $z = 0^\pm$) we get

$$\omega^2 k(\rho_2 + \rho_1) - 2k^3 B_0^2 = -(\rho_2 - \rho_1)gk^2. \quad (114)$$

Finally, solving for ω gives

$$\begin{aligned} \omega &= \left[\frac{2k^2 B_0^2}{(\rho_2 + \rho_1)} - gk\mathcal{A} \right]^{1/2} \\ &= \left[\frac{k^2 \rho_1 V_{A1}^2}{\rho_2 + \rho_1} + \frac{k^2 \rho_2 V_{A2}^2}{\rho_2 + \rho_1} - gk\mathcal{A} \right]^{1/2}, \end{aligned} \quad (115)$$

where V_{A1} and V_{A2} are the Alfvén speeds in the regions below and above the discontinuity respectively.

The term inside the square root can either be positive or negative. The term that can drive the instability is the same term that drives the hydrodynamic RT and can be simply understood as the square of the inverse of the free-fall time over a distance of $1/k$ with a modified gravity of magnitude $\mathcal{A}g$. The positive term represents the suppression of instability by magnetic tension and is the square of the frequency of a surface Alfvén wave.

12.2. Beyond 2D – the role of 3D effects and additional physics

We can now look at the dispersion relation produced from more complex settings. Extending our previous model to include perturbations that can also be perpendicular to the magnetic field leads to a small change in our dispersion relation. In this more general case, the frequency is given as [405]:

$$\omega = \left[\frac{2(\mathbf{k} \cdot \mathbf{B})^2}{\rho_2 + \rho_1} - gAk \right]^{1/2}. \quad (116)$$

The clear difference between this and the 2D case is that since the magnetic field has a direction, and perturbations perpendicular to that direction are allowed, the influence of the magnetic field is anisotropic.

We can understand more about the magnetic RT instability by thinking about two fundamental perturbations that can be applied to the system. These are the interchange mode, where $\mathbf{k} \cdot \mathbf{B} = 0$ and the undular mode where $\mathbf{k} \parallel \mathbf{B}$ (which is equivalent to the 2D setting). The interchange mode removes the influence of the magnetic field, resulting in the instability developing with the growth rate of the hydrodynamic instability at the same wavenumber. The undular mode, so called because it makes the magnetic field undulate, behaves differently. As the magnetic field will work to suppress the instability, there is a play-off between how quickly gravity can drive the instability and how quickly magnetic tension can suppress it. For this mode, simply differentiating with respect to k will reveal the k associated with the fastest growing mode:

$$k = \frac{(\rho_2 - \rho_1)g}{4B^2}. \quad (117)$$

Therefore the stronger the magnetic field strength, the smaller the wavenumber at which the most unstable undular mode forms.

Due to the fact that the undular mode includes the influence of a suppression term, for the same magnitude of k the interchange mode will always grow faster than the undular mode. The most common mode to be excited in realistic conditions is the mixed mode, where there is a component of k both parallel and perpendicular to B . For more information on the linear stage of this instability see, for example, [407].

12.2.1. Variations in the direction of the magnetic field at the interface

A well-known issue with the magnetic RT instability is that certain perturbations (for example, perturbations with $\mathbf{k} \cdot \mathbf{B} = 0$) have a growth rate which is unbounded with k . Simply put, for certain perturbations, as k goes to infinity so does the growth rate. This issue also exists for the hydrodynamic problem, where the problem is often regularized by the introduction of a region of finite size over which the density transitions, or the introduction of viscosity or surface tension.

Similarly, MHD terms can also be used to regularize the problem. This can be achieved by allowing for the magnetic field vector to be pointing in different directions above and below the density jump. Assuming a uniform magnetic field strength, the growth-rate of the instability is given by [407,408]:

$$\omega = \left[\frac{2(k_x B_x)^2 + 2(k_y B_y)^2}{\rho_2 + \rho_1} - gAk \right]^{1/2}, \quad (118)$$

where it has been assumed, without loss of generality, that the magnetic field is of the form $\mathbf{B} = [B_x, B_y, 0]$ above the discontinuity and $\mathbf{B} = [B_x, -B_y, 0]$ below (i.e. the y component of the magnetic field changes sign but not magnitude across the discontinuity whereas the x component is the same on both sides).

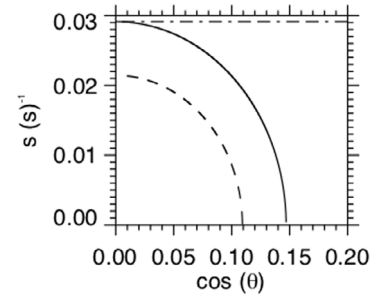


Fig. 42. Growth rate $s = i\omega$ of the uniform field (in the x direction) magnetic Rayleigh–Taylor instability (solid line), the sheared field case (dashed line) for the same field strength and the hydrodynamic case (dot-dashed line). For this calculation we have $\rho_1 = 10^{-12} \text{ kg m}^{-3}$ and $\rho_2 = 10^{-10} \text{ kg m}^{-3}$. We take a wavelength of $2 \times 10^6 \text{ m}$ and θ giving the angle between the wavevector and the x direction. In both magnetic cases, the magnitude of B is $\sqrt{\mu_0 |\mathbf{B}|} = 5 \times 10^{-4} \text{ T}$, for the sheared field case $B_y = 0.1 B_x$. For the uniform field case, only modes close to the interchange mode grow. The inclusion of the sheared field, in this case only a small amount of shear, noticeably reduces the growth rate.

It is clear from Eq. (118) that there is no longer any wavevector that can be applied to the system that removes the influence of the magnetic field. Therefore, we know that when there is magnetic shear at the density jump, magnetic tension will always work to suppress the instability, with the higher wavenumbers experiencing greater suppression. Fig. 42 shows the variation of the growth rate of the instability with the inclusion of magnetic shear, highlighting the reduction in the growth rate as a result of magnetic shear and the anisotropic influence of the magnetic field overall.

The most unstable mode is calculated by maximizing the gravitational term which drives the instability whilst minimizing the suppressive terms from the magnetic field. This can be done simply by determining which of B_x^2 or B_y^2 is smaller. Without loss of generality, we can assume that B_x^2 is the smaller. Therefore, to minimize the second term it is necessary to set k_y to be zero. Therefore, the k_x that satisfies the following equation

$$gA - \frac{4k_x B_x^2}{\rho_2 + \rho_1} = 0 \quad (119)$$

is the k_x that gives the fastest growing mode in this system [407]. This result is the same as that of the fastest growing undular mode for a uniform magnetic field of strength $|B_x|$.

12.2.2. Role of magnetic diffusion

The inclusion of diffusion has a few key effects on the dispersion relation and its derivation. To solve these systems, unfortunately the Lagrangian displacement method is not applicable, so other methods have to be employed. An example is the approach by Chandrasekhar [405] for the viscous Rayleigh–Taylor instability.

The presence of the higher order derivative in the induction equation results in there being more solutions for the spatial dependence of the eigenfunction. It can be shown that the equivalent of Eq. (111) for the diffusive case (i.e. for 2D) is

$$\left(\frac{\partial^2}{\partial z^2} - F^2 \right) \left(\frac{\partial^2}{\partial z^2} - k^2 \right) \delta B_z = 0, \quad (120)$$

where

$$F^2 = k^2 + \frac{i\omega}{\eta_D} - \frac{i\omega_A^2}{\eta_D \omega}, \quad (121)$$

and $\omega_A^2 = k^2 B_{x,0}^2 / \rho_0$. This has solutions of the form

$$\delta B_z = C_0 \exp(-kz) + C_1 \exp(kz) \quad (122)$$

$$+ C_2 \exp(-Fz) + C_3 \exp(Fz).$$

Applying $\delta B_z \rightarrow 0$ as $|z| \rightarrow \infty$ gives the solutions in each layer. We can see, because of the form of F^2 , that for large η_D , the eigenfunction behaves in exactly the same way as the ideal eigenfunction, and that in cases where ω has a real part F is complex.

The role of magnetic diffusion on the growth of the instability is that it works to remove current from the system. For a plane-wave perturbation, the larger the wavenumber along the magnetic field of the perturbation the larger the current, which means the larger the diffusive term in the induction equation. Therefore, perturbations with sufficiently large k will lose their Lorentz force, which works to suppress the instability, as diffusion will have removed all the current. The magnetic fields no longer suppress the instability at high-wavenumbers, on the contrary the fluid will no longer feel the Lorentz force and the instability can grow at the same rate as the hydrodynamic instability. However, at small k where the diffusive term is unimportant, the result will be approximately the same as the ideal MHD situation. This effect can be seen in the calculations of Díaz et al. [409], though note the diffusion used in their calculations is modified from the simple diffusion discussed here. Finally, any wave solution to the system will be subject to diffusion, so the wave should damp implying we have complex solutions giving damped waves instead of purely real frequencies as in the ideal MHD case.

12.2.3. Compressibility

An important consideration is that the investigation of the growth rate presented here has only been derived for incompressible systems. In cases where the growth rate $s = i\omega$ is such that $|s| > |\omega_{\text{comp}}|$ where ω_{comp} is the frequency of a compressible wave mode (either the slow or fast-mode in MHD), then it is likely to result in compressible effects becoming important. The result will be that work has to be done to compress the fluid instead of driving the instability, reducing its growth rate.

A simple estimate of the conditions for which compressible effects become important can be given by taking the ratio of the growth rate of the instability to the frequency of a sound wave. In situations where this ratio approaches or becomes greater than one, compressible effects are likely to become important. Taking the RTI growth rate in the incompressible limit we can calculate this ratio

$$\frac{s^2}{k^2 C_s^2} = \frac{1}{H_p k} \frac{A}{\gamma}, \quad (123)$$

where H_p is the pressure scale height defined by $H_p = k_B T / \mu_m M_p g$ with k_B the Boltzmann constant, T the temperature, μ_m the mean molecular mass and M_p is the mass of a proton. This implies that hydrodynamic RTI wavenumbers of scale $k \lesssim A/H_p$ will naturally evoke compressible effects, whereas much larger wavenumbers will be approximately incompressible.

There are two reasons why the inclusion of magnetic fields further complicates this picture. Firstly, there are two compressible wave speeds being maximized either parallel or perpendicular to the magnetic field, i.e. the slow and fast mode waves. Secondly, the growth rate of the instability depends on the angle of the wave vector to the magnetic field direction.

We can extend these considerations on hydrodynamic compressibility to that of MHD quite easily. We first note that compressible wave speeds will be smaller in the region at higher densities (which gives smaller V_A and C_s for the same magnetic field strength and gas pressure). As such, it is this layer that is most likely to be influenced by compressible effects. Therefore, the influence of compressibility is reduced for wavevectors associated with larger compressible wave speeds. For low- β plasmas this implies that wavevectors perpendicular to the magnetic field

(associated with the MHD fast-mode) will be less compressible and potentially grow faster. So compressibility is likely to create new favored directions in the system.

This was observed in the investigation by Ruderman [410], who performed a compressible extension to the incompressible study of Ruderman et al. [408]. Their study showed that compressibility does not change the critical wavevector for the instability. This can be understood as the growth rate of the instability is 0 for this perturbation, so the compressible effects modifying the instability are also 0. They also found a new favored direction in the system for instability growth, as explained above, something that does not appear in the incompressible growth rate, e.g. see Eq. (118).

12.3. The nonlinear phase of the magnetic Rayleigh–Taylor instability

Having covered some key aspects of the linear stability of the magnetic RT, we now turn our attention to the effect of the magnetic field on the nonlinear regime of RT instability.

Here, we discuss the different terms of the momentum equation that can lead to the nonlinear saturation of a magnetic RTI. Our starting point is the linear eigenfunction for the vertical Lagrangian displacement ξ_z , i.e.

$$\xi_z(\mathbf{x}, t) = \xi_z(\mathbf{0}, 0) \exp(i\mathbf{k} \cdot \mathbf{x} - k|z| - i\omega t), \quad (124)$$

where $\xi_z(\mathbf{0}, 0)$ is the value of ξ_z at $x = y = z = t = 0$. The scaling in the vertical (z) direction is $1/k$, highlighting an inherent spatial scale for the vertical displacement of the linear instability. That is to say, once the density jump has undergone a vertical deformation of distance $|\xi_z| > 1/k$, then the instability has driven the system beyond the linear regime.

We can further this argument by investigating a simple model of nonlinear development. From the equation of motion, Eq. (73), nonlinear terms exist for both the velocity and the Lorentz force. Therefore, we can expect that saturation of the instability could be caused by either of these terms. Here, we follow some of the arguments laid out in Refs. [407] and [406].

Nonlinearities in hydrodynamic systems can result in the occurrence of large shear flows that allow parasitic instabilities, e.g. the Kelvin–Helmholtz instability, to develop. In MHD systems, these saturation mechanisms are still present. If we balance the temporal derivative of the velocity with the advective derivative, we have the following scaling,

$$\frac{\partial^2 \xi}{\partial t^2} = s^2 \xi \propto s^2 \xi \cdot \nabla \xi \sim s^2 k \xi^2 \quad (125)$$

$$\Rightarrow \xi \sim \frac{1}{k}. \quad (126)$$

Therefore, it can be predicted that velocity shear created by the growth of the instability will saturate the linear stage of the instability when the boundary has been perturbed through a distance of approximately $1/k$.

Alternatively, the role of magnetic forces in halting the further evolution of the instability can be given by,

$$s^2 \xi = \frac{\partial^2 \xi}{\partial t^2} = \frac{1}{\rho_0} \nabla \times \delta \mathbf{B} \times \delta \mathbf{B} \sim \frac{k_x^2 B_0^2}{\rho_0} \xi^2, \quad (127)$$

$$\Rightarrow \xi \sim \frac{1}{k} \frac{s^2}{\omega_A^2}, \quad (128)$$

where ω_A is the frequency of a surface Alfvén wave. Therefore, when $|\omega_A| \gtrsim |s|$ we would expect that magnetic forces will be important in the nonlinear saturation of the instability.

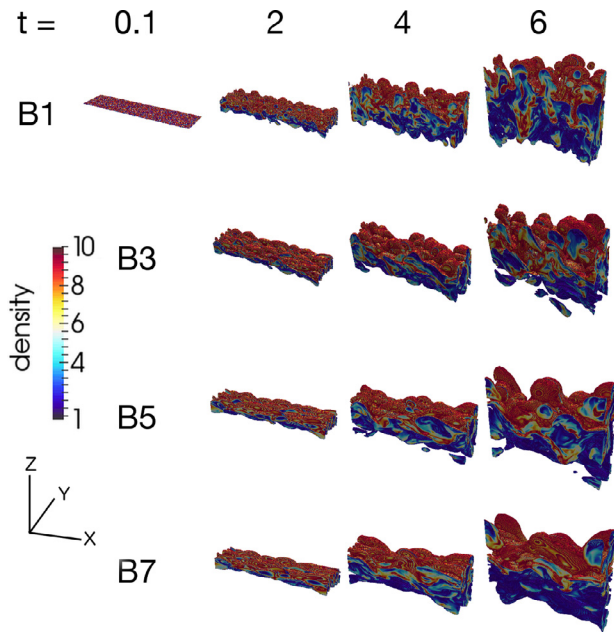


Fig. 43. Images of the development of magnetic RTI mixing layers over time for different field strengths, going from weak at the top to strong at the bottom. Fig. 2 of [411], with permission.

12.4. MHD extensions to the self-similar mixing model

In previous sections of this paper, and in many other articles [see, for example, [3]], the nonlinear evolution of the hydrodynamic Rayleigh–Taylor instability has been discussed. A key property of the nonlinear regime is that the thickness of the mixing layer grows at late time following Eq. (38), e.g. $h \approx \alpha Agt^2$. Therefore, when considering the magnetic RT instability, we are interested in (i) whether the thickness of the mixing layer grows at the late-time still scales t^2 , and (ii) whether the value of the α parameter will be different from that of the fluid turbulence cases.

To study the nonlinear magnetic RT instability, numerical simulations are generally employed. The particular case of nonlinear mixing has thus far been addressed by few groups, with studies by Stone & Gardiner [412,413], and Carlyle et al. [414]. A set of simulations of the mixing process is shown in Fig. 43, taken from [411], where the effect of increasing the field strength on the development of the mixing layer is investigated. The simulation with the weakest field strength is shown at the top with progressively stronger fields further in the images below. It is clear that the increase in field strength results in the creation of elongated structures along the magnetic field due to the increasing significance of magnetic tension.

Using these simulations, the value of α as it depended on the strength of the magnetic field (in this case the non-dimensional parameter J_0 , defined as $J_0 \equiv V_A^2/gL$, was used) has been calculated. The α values found in these simulations (~ 0.04) [412–414] are larger than the hydrodynamic value of $\alpha \sim 0.025$ found in comparable simulations [412,413]. At the lower J_0 values ($J_0 \sim 0.03$), a suggestion for the increase in α compared to hydrodynamic mixing is the suppression of secondary Kelvin–Helmholtz instability by the magnetic field. This increases the efficiency with which the boundary can be distorted as suggested by Stone & Gardiner [412,413]. However, increasing the field strength further results in a decrease in the α value, but it is unclear if α ever approaches the hydrodynamic value. It is clear that there are many open questions in this area.

One question that needs to be answered still is if mixing in MHD flows has a self-similar phase or if the changing importance of gravity against the magnetic field over lengthscale (as seen in the linear dispersion relation) means that the mixing is fundamentally a function of system size? If a broadband perturbation were imposed on the system, large wavelengths along the magnetic field would grow, and would be likely to be associated with the thicker mixing layers. This would imply that the J_0 parameter varied with layer width, so that α could also be expected to vary with the width. Also, the corresponding evolution when narrowband perturbations are imposed is a question that still remains to be investigated.

In this section we have looked at a case when a coherent horizontal magnetic field influences the mixing evolution. Another possible case is where a very weak magnetic field is enhanced through the turbulence driven by the RT instability (this would be in a regime where the instability behaved in a fluid like way). This possibility is discussed in Section 13.3

13. The magnetic Richtmyer–Meshkov instability

13.1. Shock waves

Like gases, plasmas support shock waves. The conditions on the shock waves can be determined in a manner similar to the corresponding gas dynamic case. For a surface of discontinuity in MHD, the conservation properties of (72)–(76) (neglecting gravity) yield [19],

$$\begin{aligned} \llbracket \rho u_n \rrbracket &= 0, \quad \llbracket B_n \rrbracket = 0, \\ \rho u_n \llbracket \mathbf{u}_t \rrbracket &= B_n \llbracket \mathbf{B}_t \rrbracket \\ \rho u_n \left[\left[\frac{\mathbf{B}_t}{\rho} \right] \right] &= B_n \llbracket \mathbf{u}_t \rrbracket, \quad \left[\left[\rho u_n^2 + p + \frac{1}{2} B_t^2 \right] \right] = 0, \\ \rho u_n \left[\left[\frac{1}{2} (u_n^2 + u_t^2) + \frac{p}{(\gamma - 1)\rho} + \frac{p}{\rho} + \frac{B_t^2}{\rho} \right] \right] &= B_n \llbracket \mathbf{u}_t \cdot \mathbf{B}_t \rrbracket. \end{aligned} \quad (129)$$

In order, these state conditions for conservation of mass, solenoidal magnetic field, and conservation of tangential momentum, tangential magnetic flux, normal momentum, and energy.

While the character of MHD shock waves is thus derived in a similar way to that of their gas dynamic counterparts, the situation here is much richer. This is because in non-conducting gases, information is transmitted throughout the fluid only through sound waves, and the shock Mach number which is defined in terms of the sound speed characterizes the shock jump conditions. That is, the jumps (A.2) are controlled by only one parameter, which we usually define as the Mach number. By contrast, there are three types of waves that can travel through MHD fluids; these are the Alfvén wave, and the fast and slow magnetosonic waves (see Section 11 for more details), each traveling at a different characteristic speed; thus there are three parameters that control the jump conditions (129). Hence, we can define three different Mach numbers for the MHD shock in terms of each speed, say M_f, M_l, M_s respectively, the l representing the *intermediate* wave speed. We thus encounter a situation where, in the shock frame, six different configurations are permitted by the jump conditions. Assuming the characteristic speeds are distinct, these configurations are shown in Table 1, where F, S and I designate the *fast* magnetosonic, *slow* magnetosonic, and *intermediate* shock types respectively, and the numbering indicates where in the ordering of characteristic speeds the upstream and downstream velocities in the shock frame are placed.

Each of these shocks has its own unique character. The intermediate shocks in particular are controversial in the literature,